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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

August 2023

22MCA201 - DATA COMMUNICATION AND NETWORKS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What is the design goal of scrambling technique? Explain two scrambling techniques with examples.	10	L2	CO1	PO2
	b)	Compare and contrast the OSI model and TCP/IP protocol architecture.	10	L1	CO1	PO1
2.	a)	Encode 10101110001 bits using i) Bipolar AMI ii) Manchester iii) Differential Manchester iv) NRZ-I v) PSK	5 5	L3 L1	CO1 CO1	PO2 PO1
	b)	Explain a simple communication model with the help of a diagram.	5	L1	CO1	PO1
	c)	Compute CRC for given data M=1101001011 P=11011 using polynomials (i) Use polynomial long division to determine the message that should be transmitted. (ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?	10	L3	CO1	PO2
		Unit – II				
3.	a)	Write a note on : i) MIME ii) SMTP iii) GO-BACK-N ARQ. iv) POP3 v) IMAP	10	L1	CO2	PO2
	b)	Explain a frame structure of HDLC and include a diagram to illustrate its operation.	10	L1	CO2	PO2
4.	a)	Differentiate between flow control and error control. Explain various techniques of flow control with examples.	10	L2	CO2	PO1
	b)	Write a note on Domain Name Space and Name Servers.	10	L1	CO2	PO1
		Unit – III				
5.	a)	Discuss the different TCP socket Primitives.	10	L1	CO3	PO1
	b)	Describe the elements of transport protocols, including addressing, connection establishment, and connection release.	10	L1	CO3	PO1
6.	a)	Explain the concept of Remote Procedure Call (RPC) and discuss its role in facilitating communication between networked applications using UDP.	10	L2	CO3	PO1
	b)	Describe the fields of the TCP segment Header with suitable diagram.	10	L2	CO3	PO1
		Unit – IV				
7.	a)	Explain the concepts of distance vector routing and link state routing algorithms, discuss their differences.	10	L1	CO4	PO1
	b)	Compare Virtual circuit subnet with datagram subnet with a suitable example.	10	L2	CO4	PO1
8.	a)	Write a note on i) Store-and-Forward Packet Switching ii) Optimality Principle	10	L1	CO4	PO1
	b)	Explain the concepts of multi-cast Routing and Broadcast Routing algorithms.	10	L1	CO4	PO1

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Unit – V

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| 9. | a) | Explain the IPv6 header format, including its fields and their purposes, with the help of a diagram, and also Compare the IPv6 header format with the IPv4 header format, highlighting the key differences. | 10 | L2 | CO5 | PO1 |
| | b) | Describe the concept of load shedding in congestion control and explain how it helps alleviate network congestion. | 10 | L2 | CO5 | PO1 |
| 10. | a) | Discuss the techniques of achieving good quality of service. | 10 | L2 | CO5 | PO1 |
| | b) | Differentiate between leaky bucket and token bucket algorithm. Describe both the algorithms with a neat diagram. | 10 | L2 | CO5 | PO1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
